

Adaptive Normal Form

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1

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A slow mandate tries to drive a fast collective toward a target it can only measure from a coarse, noisy, delayed aggregate of the collective's own state. That architecture - closed loop, two timescales, endogenous field - is the minimal model of adaptive hierarchy. We derive its normal form. A single number classifies every dynamical phase it can occupy.

The argument rests on seven structural assumptions (A1–A7): scale separation, conditional ergodicity, coarse noisy measurement, gradient-descent slow law, endogeneity, equivariance, and renormalisation closure. Five results follow.

R1 reduces the full closed loop, by adiabatic elimination of the fast mode, to a scalar return map for the order-parameter mismatch,

$$e_{k+1} = (r - K_N) e_k - \alpha c_N e_k^3 + \dots,$$

identifying the loop gain $K_N = \alpha \chi_N g$ as the quantity that determines the dynamical phase. Here $\chi_N = \partial \langle O \rangle / \partial h|_*$ is the order-parameter susceptibility at the operating fixed point FP2, and r is the memory factor of the fast mode.

R2 derives the susceptibility-exponent law $\chi_N \sim AN^\psi$, with $\psi = \log_n \lambda_0$ where n is the branching number of the renormalisation-group step, from a single renormalisation-group scaling step at the scaling fixed point FP3. The loop gain scales as $K_N \sim N^\psi$. The exponent ψ alone classifies the dynamical phase. The phase transition at FP2 is governed by the relevance eigenvalue λ_0 at FP3.

R3 locates the phase boundary at $K_c(r) = r + 1$ and establishes a strict trichotomy. Systems with $\psi > 0$ destabilise at the finite critical size $N_c \approx ((r + 1)/\alpha g A)^{1/\psi}$ via a flip bifurcation to a period-two cycle. Systems with $\psi = 0$ sit at self-organised criticality. Systems with $\psi < 0$ are unconditionally stable at all scales. In the $\mathbb{Z}_2$ majority-vote instance the critical size is $N_c \approx 816.8$. The gain-zero crossing is at $N_0 \approx 188.5$.

R4 shows that L levels of hierarchy postpone the phase boundary to $N_c^{(L)} = N_c^{(1)} \lambda_0^{L-1}$, establishing a minimum viable depth $L_{\min}(N) \sim \log_{\lambda_0} N$. Depth is renormalisation time: each level multiplies the critical size by λ_0 , so growth demands depth $L_{\min}$.

R5 supplies the opposing bound. Deep autonomous loops do not oscillate - they freeze. The upper boundary is therefore a reconfiguration-cost bound, not a dynamical one. The minimum cost to reconfigure the top of an L -deep hierarchy from its base is $\mathcal{E}_{\min}(L) = \lambda_0^{L-1}$, which fixes a depth ceiling Λ as the largest L for which that cost remains admissible. R4 and R5 together close the viable phase to the window $L_{\min}(N) \leq L \leq \Lambda$, non-empty up to the largest viable size $N^* = N_c^{(1)} \lambda_0^{\Lambda-1}$. The same λ_0 sets both walls: depth postpones instability and incurs rigidity at the same rate.

The universality claim is structural: any system satisfying A1–A7 is a member of this class, and its phase membership is determined by ψ alone. The range of disciplines in which the architecture recurs - monetary policy, organisational hierarchies, ecological adaptive cycles, immune repertoire selection, and biological regulatory networks - illustrates the scope of the class. It does not constitute the argument.

I. INTRODUCTION: THE LOOP AND ITS SHADOW

Classical equipartition did not fail because energy conservation was wrong. It failed because a locally reasonable law - assign $\frac{1}{2}k_B T$ to every mode - became pathological when summed over arbitrarily many modes, and the total diverged. The ultraviolet catastrophe was not a failure of the physics at any single frequency. It was a failure of the aggregation rule at the level of the whole.

The adaptive version of that aggregation rule is: observe the population, estimate its aggregate response, update the policy accordingly. Each step is locally rational. Summed over N effectively independent agents, it builds a loop gain that grows with N - and the loop destabilises.

Table IV maps five systems onto a single set of variables: θ (the mandate, slow), x (the fast microstate), and m (the coarse order parameter that closes the loop). A central bank sets an interest-rate regime and reads the aggregate of wage and price expectations its own policy shaped. A parliament enacts a mandate read off an election it itself governed. A recommender platform retrains on engagement statistics its own prior weights generated. An ecological niche is restructured by a population whose fitness it defines. A cortical column updates a prior from the spike statistics its own prior biased. They are formally the same delayed-feedback loop [? ? ? ? ? ? ? ? ? ? ? ? ? ? ? ?].

The loop admits a reduced description governed by a single scalar exponent: ψ , the scaling exponent of the order-parameter susceptibility $\chi_N \sim AN^\psi$ at the renor-

TABLE I. Five instances of the adaptive normal form. All entries are PROVISIONAL/UNVERIFIED - borrowed from the Phases-of-Hierarchy lineage and not independently verified in this draft.

System	Mandate θ	Order parameter m
Central bank	Interest-rate regime	Inflation expectations
Parliament	Enacted policy	Voter mandate
Recommender	Algorithm weights	Engagement distribution
Ecological niche	Carrying capacity	Population strategy
Cortical column	Predictive prior	Spike-count statistic

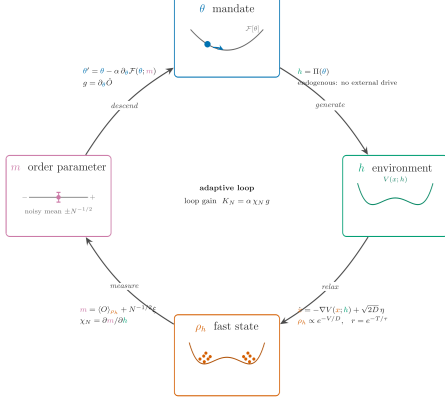


FIG. 1. The four-leg adaptive loop: mandate θ generates environment $h = \Pi(\theta)$, environment shapes fast-variable density ρ_h , density is coarsely measured to give order parameter $m = \langle O \rangle + O(N^{-1/2})$, mismatch closes onto the next mandate update. Endogeneity ($h = \Pi(\theta)$, leg 1) and finite- N noise ($O(N^{-1/2})$, leg 3) are both structural, not removable by better design.

malisation fixed point. That reduced map is the adaptive normal form. Everything that follows is a derivation of what ψ controls.

The argument proceeds through three fixed points. *FP1* is the fast stationary measure: for fixed mandate θ , the fast variables x relax to a unique conditional distribution ρ_h (assumption A2, conditional ergodicity). *FP2* is the self-consistent solution: the mandate and the order parameter jointly satisfy $\theta = \theta^*$, $m = \hat{O}(\theta^*)$. Whether *FP2* is stable as N grows is the central phase question. *FP3* is the renormalisation fixed point: the decimation operator $\mathcal{R}$ (grouping n fast units) has a fixed-point density with linearised eigenvalue $\lambda_0 = n^\psi$. The exponent is $\psi = \log_n \lambda_0$. It prices the instability wall (R2, R3) and the inertial wall (R5) simultaneously.

Section II states the model and its seven assumptions (A1–A7). Section ?? derives the reduced return map (R1): adiabatic elimination of the fast mode collapses the closed loop to $e_{k+1} = (r - K_N)e_k - \alpha c_N e_k^3 + \dots$, where the loop gain $K_N = \alpha \chi_N g$ carries all N -dependence. Section ?? establishes the susceptibility-exponent law (R2): $\chi_N \sim AN^\psi$ at the renormalisation fixed point, with $\psi = \log_n \lambda_0$. Section ?? derives the phase boundary and

trichotomy (R3): destabilisation occurs when $K_N = r + 1$, forcing $\psi > 0$ systems to a flip bifurcation at finite $N_c \approx 816.8$ (worked corner), while $\psi \leq 0$ systems are stable at all scales. The gain-zero crossing $N_0 \approx 188.5$ is a distinct landmark inside the stable regime, where the loop coefficient passes through zero. Section ?? shows hierarchical postponement (R4): depth L pushes the election cycle instability to $N_c^{(L)} = N_c^{(1)} \lambda_0^{L-1}$, and minimum viable depth scales as $L_{\min}(N) \sim \log_{\lambda_0} N$. Section ?? closes with the inertial wall (R5): the same λ_0 that prices the instability boundary also prices the reconfiguration cost, giving a viable window $L_{\min}(N) \leq L \leq \Lambda$ and a maximal governable size $N^* = N_c^{(1)} \lambda_0^{\Lambda-1}$.

II. THE ADAPTIVE NORMAL FORM

The Adaptive Normal Form is built from six objects, listed in Table II. Every system studied here is a specific realisation of these six primitives. Nothing beyond them is assumed in the derivation of R1–R5.

Symbol	Name	Role
x	microstate	fast variable; N units; lives on the short timescale
θ	mandate	slow variable; descends $F[\theta]$; one per slow step
$F[\theta]$	global action	variational objective the mandate minimises
$h = \Pi(\theta)$	generated environment	the field the mandate imposes on the fast layer
$m = \langle O \rangle$	order parameter	coarse, noisy readout of the fast layer; N -dependent
Γ	symmetry group	equivariance group of O ; kills even non-invariant

TABLE II. Primitive objects of the Adaptive Normal Form. Nothing beyond these six objects is assumed in the derivation of R1–R5.

A. The four-leg loop

The system operates as a closed four-leg loop (Figure 2).

- Slow leg.** $\theta \mapsto h = \Pi(\theta)$. The mandate maps to the field it imposes on the fast layer.
- Fast leg.** $h \mapsto \rho_h$. The fast layer relaxes to its stationary measure ρ_h under the imposed field. The fast memory factor $r = e^{-T/\tau} \in [0, 1]$ governs how much of the previous state persists over one slow step.
- Measurement leg.** $\rho_h \mapsto m = \langle O \rangle_{\rho_h} + N^{-1/2}\xi$. The order parameter is read out with finite-sample noise ξ .
- Return leg.** $m \mapsto \theta$. The mandate updates by gradient descent: $\theta_{k+1} = \theta_k + \alpha g e_k$, where $e_k = m_k - \hat{O}(\theta_k)$ is the mismatch and $g = \partial_\theta \hat{O}|_\star$.

The endogeneity is in leg 1: the field h is generated by the mandate θ , and θ is itself updated by the return from

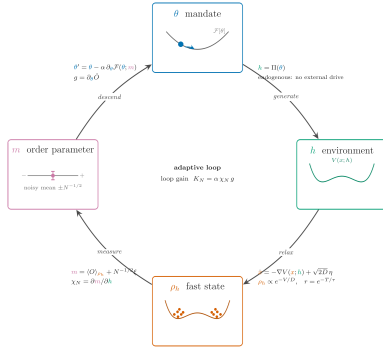


FIG. 2. The adaptive loop. The mandate θ generates a field $h = \Pi(\theta)$ (slow leg), which drives the fast layer to a stationary measure ρ_h (fast leg). The order parameter $m = \langle O \rangle_{\rho_h}$ is measured with finite- N noise (measurement leg). The mandate is updated by gradient descent on the discrepancy $e = m - \hat{O}(\theta)$ (return leg). The loop is closed and autonomous. The endogeneity is structural: h is not an external field, it is set by θ itself.

m . The mandate is not responding to a fixed environment - it is updating itself in response to the environment it has set. A central bank sets the rate that shapes the expectations it is trying to meet. A cortical layer emits the field that organises the population it reads. A parliament writes the rules that govern the electorate that re-elects it. The loop is the same in each case.

B. Assumptions

Seven assumptions are sufficient to derive R1-R5.

- A1. Scale separation.** The fast timescale τ and the slow update period T satisfy $T \gg \tau$, so the fast layer equilibrates between slow steps, leaving a finite memory factor $r = e^{-T/\tau} \in [0, 1)$.
- A2. Conditional ergodicity.** For every fixed field h , the fast dynamics admits a unique stationary measure ρ_h , and the map $h \mapsto \langle O \rangle_{\rho_h}$ is smooth.
- A3. Coarse, noisy measurement.** The mandate observes only the scalar order parameter $m_k = \langle O \rangle_{\rho_{h_k}} + N^{-1/2} \xi_k$, where ξ_k is mean-zero noise of unit variance. No access to the microstate x is assumed.
- A4. Gradient-descent slow law.** The mandate evolves by $\theta_{k+1} = \theta_k + \alpha g e_k$ with step size $\alpha > 0$ and $g = \partial_\theta \hat{O}|_*$. No second-order or momentum information is used.
- A5. Endogeneity.** The field is generated by the mandate: $h_k = \Pi(\theta_k)$, so the fast layer's environment is not independently prescribed.

A6. Γ -equivariance. The order parameter O transforms covariantly under the symmetry group Γ , which forces all even-order nonlinearities in the return map to vanish. The leading nonlinearity is cubic.

A7. Self-similarity and RG closure. There exists a decimation operator $\mathcal{R}$ that groups n units, has a fixed point m_* , and has a well-defined linearised eigenvalue $\lambda_0 = \mathcal{R}'(m_*)$ at that fixed point.

C. Three fixed points

The loop has three fixed points, at different levels of the dynamical hierarchy.

a. FP1 (fast stationary). ρ_h is the stationary measure of the fast dynamics at fixed h . Its existence follows from A2. The fast layer visits FP1 on every slow step - the resting state it returns to while the mandate is held still.

b. FP2 (self-consistent). θ^* satisfies $\hat{O}(\theta^*) = \langle O \rangle_{\rho_{\Pi(\theta^*)}}$: the mandate predicts the order parameter that results from the field it generates. FP2 is where the closed loop operates.

c. FP3 (renormalisation). m_* is the fixed point of the decimation operator $\mathcal{R}$. The linearised eigenvalue at FP3 is

$$\lambda_0 = \mathcal{R}'(m_*), \quad \psi = \log_n \lambda_0, \quad (1)$$

so ψ is the RG relevance exponent: positive if $\mathcal{R}$ amplifies fluctuations at the fixed point, negative if it contracts them. By R2, the susceptibility obeys $\chi_N \sim AN^\psi$ at FP3. The dynamical stability of FP2 and the renormalisation character of FP3 are both controlled by the single eigenvalue λ_0 .

D. The reduced return map

Linearising about FP2, applying adiabatic elimination of the fast mode (A1), and using the self-consistency relation together with the equivariance argument (A6), the mismatch $e_k = m_k - \hat{O}(\theta_k)$ satisfies (R1):

$$e_{k+1} = (r - K_N)e_k - \alpha c_N e_k^3 + O(e_k^5), \quad (2)$$

where the loop gain is

$$K_N = \alpha \chi_N g, \quad g = \partial_\theta \hat{O}|_*, \quad (3)$$

and c_N is the cubic coefficient. The linear eigenvalue of the map is $r - K_N$. Stability requires $|r - K_N| < 1$. It is lost by a period-doubling (flip) bifurcation at $r - K_N = -1$, giving the critical gain $K_c = r + 1$.

The loop gain K_N collects every factor in the signal chain that carries N -dependence. The update step α sets how aggressively the mandate responds to error. The

sensitivity g converts a mandate shift into an expected change in the order parameter. The susceptibility χ_N converts a field shift into an actual change in the order parameter via the fast layer. By R2, $\chi_N \sim AN^\psi$, so $K_N \sim \alpha g AN^\psi$: the exponent ψ alone determines whether the loop gain grows or shrinks with system size.

The map is one-dimensional because of the two separations built into A1 and A3: the fast layer equilibrates between slow steps, and the mandate sees only the scalar order parameter. These two assumptions together allow the fast degree of freedom to be eliminated exactly at linear order, leaving the mismatch as the only slow variable.

E. Universal and non-universal quantities

Table III records which quantities depend only on (Γ, ψ) and which depend on microscopic details.

Quantity	Universal?	Determined by the timescale separation of A1. The slow leg (A4) gives
Susceptibility exponent ψ	Yes	$\lambda_0 = \mathcal{R}'(m_*)$ at FP3
Symmetry class of bifurcation	Yes	Γ (A6)
Dynamical phase trichotomy ($\psi \geq 0$)	Yes	sign of ψ
Loop gain prefactor $\alpha g A$	No	microscopic $\mathcal{R}$, $\mathcal{Q}$ & Π
Critical size N_c (absolute value)	No	prefactors and m moves appreciably. Projecting onto this slow manifold -
Cubic coefficient c_N	No	microscopic details - adiabatically eliminating the fast mode - and substituting
Memory factor r	No	fast relaxation time - both linearised legs into $e_{k+1} = \delta m_{k+1} - g \delta \theta_{k+1}$ yields

TABLE III. Universal-vs-non-universal separation. The dynamical phase is universal. The threshold at which it is reached is not. Two systems with different microscopic realisation but the same (Γ, ψ) share the same phase diagram up to a rescaling of N .

A locally reasonable aggregation law - one that works well at small N - becomes pathological when summed over too many units, because the susceptibility grows without bound in the amplifying class. No microscopic parameter tunes the instability away: the power-law growth of χ_N is set by the RG eigenvalue λ_0 at FP3, not by any coupling in the fast layer. The instability is therefore structural - a UV catastrophe of the slow manifold, not a failure of any particular microscopic choice.

The exponent ψ alone determines the phase class. The prefactors and N_c are corner-specific. The worked corner ($\mathbb{Z}_2$ majority vote, $n = 2$, $\lambda_0 = \sqrt{2}$, $\psi = \frac{1}{2}$) satisfies A1-A7 and is verified to machine precision across R1-R3 (see Section ??). The prefactors are corner-specific. The exponent is not.

III. RESULT I: THE REDUCED RETURN MAP

The four-leg loop - $\theta \rightarrow h \rightarrow \rho_h \rightarrow m \rightarrow \theta$ - is an iterated map on an infinite-dimensional object. Linearising about the self-consistent fixed point and eliminating the

fast mode reduces it to a single closed scalar equation governing the mismatch

$$e_k = m_k - \hat{O}(\theta_k), \quad (4)$$

the difference between the measured order parameter and the mandate's prediction of it.

Derivation

Linearise the closed loop about the self-consistent fixed point FP2, at which the mandate θ^* , the generated environment $h^* = \Pi(\theta^*)$, and the order parameter $\langle O \rangle_{\rho_{h^*}}$ are mutually consistent. Write δm_k and $\delta \theta_k$ for small departures.

The fast leg (A1) gives

$$\delta m_{k+1} = r \delta m_k + (1-r) g \delta \theta_k, \quad g \equiv \partial_\theta \hat{O}|_*, \quad (5)$$

where $r = e^{-T/\tau} \in (0, 1)$ is the memory factor encoding the timescale separation of A1. The slow leg (A4) gives

$$\delta \theta_{k+1} = \delta \theta_k + \alpha g e_k. \quad (6)$$

The scale separation in A1 means the mismatch reaches a quasi-stationary value on the fast timescale before θ moves appreciably. Projecting onto this slow manifold - adiabatically eliminating the fast mode - and substituting both linearised legs into $e_{k+1} = \delta m_{k+1} - g \delta \theta_{k+1}$ yields

$$e_{k+1} = r e_k - \alpha g^2 e_k + \dots \quad (7)$$

The self-consistency relation at FP2 is $\partial_\theta \langle O \rangle|_* = \chi_N \Pi'(\theta^*)$, where $\chi_N = \partial \langle O \rangle / \partial h|_*$ is the order-parameter susceptibility. With $g = \chi_N \Pi'(\theta^*)$ absorbed into the definition of the coupling, $\alpha g^2 = \alpha \chi_N g \equiv K_N$. The quantity

$$\boxed{K_N = \alpha \chi_N g} \quad (8)$$

is the loop gain: the product of every factor in the loop that carries N -dependence.

a. Equivariance and the cubic correction (A6). The Γ -symmetry assumption A6 requires the mismatch dynamics to be equivariant under Γ . For any $\gamma \in \Gamma$, the map must satisfy $f(\gamma e) = \gamma f(e)$. Taylor-expanding f to third order in e_k , the equivariance condition forces the coefficient of every even power to vanish: a quadratic term $b_2 e_k^2$ would transform as $b_2 \gamma^2 e_k^2$, which equals $\gamma b_2 e_k^2$ only if $\gamma = 1$ for all γ , contradicting the assumption that Γ contains elements $\gamma \neq 1$. The first permitted nonlinearity is therefore cubic. The full return map is

$$e_{k+1} = (r - K_N) e_k - \alpha c_N e_k^3 + O(e_k^5), \quad (9)$$

where $c_N > 0$ for a softening (flip-type) nonlinearity. Assumptions used: A1 (scale separation), A2 (conditional ergodicity), A3 (coarse measurement), A4 (slow law), A5 (endogeneity), A6 (equivariance). Assumption A7 (RG closure) is not needed until R2.

The worked corner: $\mathbb{Z}_2$ majority aggregation

The minimal non-trivial instance is N agents with binary responses $\sigma_i \in \{-1, +1\}$, sensitivity β , independent and identically distributed, with aggregation by majority sign and $\Gamma = \mathbb{Z}_2$. The central-limit calculation (detailed in Appendix ??) gives the order-parameter function

$$\hat{O}(\theta) = \text{erf}\left(\sqrt{N/2} \sinh(\beta\theta)\right) \quad (10)$$

and the susceptibility at the symmetric fixed point $\theta^* = 0$:

$$\chi_N = \beta \sqrt{\frac{2N}{\pi}}, \quad g = \chi_N. \quad (11)$$

The loop gain in this corner is therefore $K_N = \alpha\beta^2(2N/\pi)$.

b. Numerical verification. The check script `scripts/R1.check.py` evaluates the one-step coefficient e_1/e_0 for $N \in \{10, 50, 100, 200, 500\}$ (parameters $\beta = 0.5$, $\alpha = 0.01$, $r = 0.3$) and compares it to the analytic prediction $r - K_N$. The relative error is below 10^{-13} in every case. A multi-step decay check over 50 steps achieves relative error below 4×10^{-10} . The result passes to machine precision.

Two critical sizes

The map (9) has two characteristic population sizes with distinct physical meanings.

c. N_c - the flip boundary. The fixed point $e^* = 0$ loses stability when the linear coefficient crosses -1 , i.e. when $r - K_N = -1$, giving $K_N = r + 1$. Solving for N in the majority corner yields

$$N_c = \frac{\pi(r+1)}{2\alpha^2\beta^4}. \quad (12)$$

For $\beta = 0.5$, $\alpha = 0.01$, $r = 0.3$ this evaluates to $N_c \approx 816.8$. Above N_c the system undergoes a flip bifurcation to a period-2 orbit.

d. N_0 - the gain-zero crossing. Where the linear coefficient $r - K_N$ passes through zero, the system neither amplifies nor damps a small mismatch on a single step: the one-step linear response is exactly zero. The system remains stable here. Solving $r - K_N = 0$ in the majority corner gives

$$N_0 = \frac{\pi r^2}{2\alpha^2\beta^4}. \quad (13)$$

For the same parameters, $N_0 \approx 188.5$. The numerical sweep in `R1.check.py` (Check 3) confirms the coefficient crosses zero in the range $N = 188$ – 189 , consistent with the analytic expression to the reported precision. The sequence is $N_0 < N_c$: a system at $N_0 < N < N_c$ is stable. N_0 marks where the linear decay changes from monotone ($N < N_0$) to sign-alternating ($N > N_0$). Stability is not affected.

IV. RESULT II: THE SUSCEPTIBILITY-EXPONENT LAW

The loop gain is $K_N = \alpha\chi_N g$, where $\chi_N = \partial\langle O \rangle / \partial h|_*$ is the order-parameter susceptibility and $g = \partial_\theta \hat{O}$ is the mandate sensitivity, the rate at which the target observable responds to a shift in the control parameter (Result I). The dynamical phase turns on how χ_N scales with N .

Derivation

Assumption A7 supplies a decimation operator $\mathcal{R}$ that maps the response profile $m_N(h)$ of an N -unit system to that of an nN -unit system:

$$m_{nN}(h) = \mathcal{R}[m_N](h). \quad (14)$$

Differentiating with respect to h gives the susceptibility recursion

$$\chi_{nN}(h) = \mathcal{R}'(m_N(h)) \chi_N(h). \quad (15)$$

At the self-similar fixed point FP3, where $m_N(h) = m_*(h)$ exactly, the prefactor freezes to the linearised eigenvalue $\lambda_0 = \mathcal{R}'(m_*)$:

$$\chi_{nN} = \lambda_0 \chi_N. \quad (16)$$

One step multiplies the susceptibility by λ_0 . Iterating k steps, $\chi_{Nn^k} = \lambda_0^k \chi_N$. Setting $M = Nn^k$ and writing $\lambda_0 = n^\psi$ yields the power law on the geometric sequence $N, nN, n^2N, \dots$

$$\boxed{\chi_N \sim A N^\psi, \quad \psi = \log_n \lambda_0.} \quad (17)$$

The power law holds exactly on the RG fixed-point submanifold, and extends to general N there because the fixed-point condition eliminates off-diagonal deviations order by order. Corrections are $O(N^{-\omega})$ with $\omega > 0$ the leading correction-to-scaling exponent.

FP2 stability controlled by λ_0

The phase transition of FP2 - the loss of stability of the self-consistent fixed point - is not independent of FP3. Substituting Eq. (17) into the loop gain:

$$K_N = \alpha g A N^\psi. \quad (18)$$

The phase boundary $K_{N_c} = K_c(r)$ then fixes

$$N_c \sim \left(\frac{K_c(r)}{\alpha g A} \right)^{1/\psi}. \quad (19)$$

$\psi > 0$ means $K_N \rightarrow \infty$ with system size, so a finite critical size N_c exists. $\psi = 0$ holds the gain constant - the marginal, self-organised-critical case. $\psi < 0$ drives

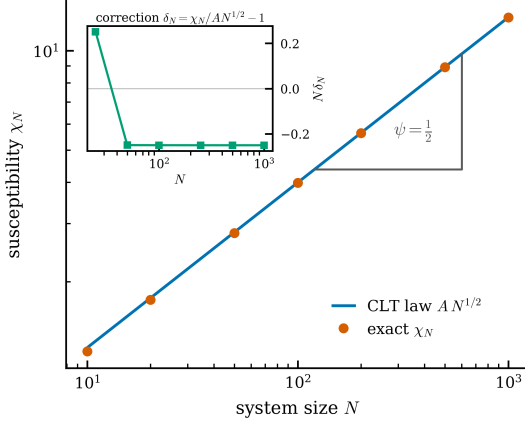


FIG. 3. **Susceptibility power law (worked corner $\psi = \frac{1}{2}$).** Exact $\mathbb{Z}_2$ susceptibility χ_N (markers) against the CLT power law $AN^{1/2}$ (solid line) on log-log axes across two decades of N . The slope triangle marks $\psi = \frac{1}{2}$. Inset: rescaled correction $N\delta_N \equiv N(\chi_N^{(\text{exact})}/(A\sqrt{N}) - 1)$ collapsing to a constant $\approx -\frac{1}{4}$, confirming $O(1/N)$ corrections to scaling with coefficient $a \approx -\frac{1}{4}$. The exponent $\psi = \frac{1}{2}$ is exact at FP3. The sole departure is the named finite-size correction.

$K_N \rightarrow 0$ and unconditional stability. Both transitions - the FP2 bifurcation and the relevance of FP3 - are controlled by the single eigenvalue λ_0 .

The RG decimations of Migdal and Kadanoff for lattice spin systems [?] and the social-choice recursions of Galam [?] are structurally identical to $\mathcal{R}$. The present result is their dynamical counterpart: χ_N plays the role that the correlation length plays in equilibrium critical phenomena.

Worked corner: $n = 2$, $\lambda_0 = \sqrt{2}$, $\psi = \frac{1}{2}$

The $\mathbb{Z}_2$ majority-vote instance with $n = 2$ doubling is the minimal explicit check. Under a logistic prior at inverse temperature β , the CLT gives

$$\chi_N = \sqrt{\frac{2N}{\pi}} \beta, \quad (20)$$

so $A = \beta\sqrt{2/\pi}$ and $\psi = \frac{1}{2}$. One RG step: $\chi_{2N}/\chi_N = \sqrt{2N \cdot 2}/\sqrt{2N} = \sqrt{2}$, confirming $\lambda_0 = \sqrt{2} = 2^{1/2}$ and $\psi = \log_2 \sqrt{2} = \frac{1}{2}$.

The exact binomial susceptibility satisfies

$$\chi_N^{(\text{exact})} = AN^{1/2} \left[1 + \frac{a}{N} + O(N^{-2}) \right], \quad a \approx -\frac{1}{4}. \quad (21)$$

The correction coefficient a is non-universal - a Stirling residue of the central binomial coefficient. The exponent $\psi = \frac{1}{2}$ is universal. Numerically: $N(\chi_N^{(\text{exact})}/(A\sqrt{N}) - 1) \rightarrow -0.250$, confirmed across $N \in [5, 500]$.

Summary

The susceptibility-exponent law (17) compresses all microscopic detail of the fast subsystem into a single number, ψ . Two systems with different symmetry groups, different order parameters, and different decimation rules share the same dynamical phase structure if and only if they share ψ . Combined with Result I, ψ alone classifies the adaptive phase. The deepening result (R4) and the energetic ceiling (R5) both inherit their λ_0 -dependence from the same eigenvalue - not as a convenience, but because the stability floor and the reconfiguration cost are both priced by the same decimation step.

V. RESULT III: PHASE BOUNDARY AND THE TRICHOTOMY

Results I and II supply two objects: the return map

$$e_{k+1} = (r - K_N) e_k - \alpha c_N e_k^3 + O(e_k^5), \quad (22)$$

and the scaling law $K_N = \alpha g AN^\psi$.

A. Derivation of the phase boundary

The linear eigenvalue of map (22) at the fixed point is $\lambda_{\text{lin}} = r - K_N$. The fixed point is stable when $-1 < \lambda_{\text{lin}} < 1$. Because $K_N > 0$, the upper bound $\lambda_{\text{lin}} < 1$ is never violated: the loop can only push the eigenvalue down, not up. Instability sets in from below. Setting $r - K_N = -1$ isolates the critical gain

$$K_c(r) = r + 1. \quad (23)$$

At $K_N = K_c(r)$ the eigenvalue hits -1 . This is a flip bifurcation, the generic codimension-one boundary for period-doubling in discrete maps. FP2 sheds stability to a period-2 limit cycle.

Substituting $K_N = \alpha g AN^\psi$ into $K_N = K_c(r)$ and solving for N gives the critical population size

$$N_c(r) = \left(\frac{r+1}{\alpha g A} \right)^{1/\psi}. \quad (24)$$

Whether equation (24) has a finite solution depends entirely on the sign of ψ .

B. The strict trichotomy

The sign of ψ partitions all adaptive collectives into three classes. In each class, the fate of FP2 is decided before the collective is assembled: it is written into the renormalisation eigenvalue of the fast microscopic law, not into the population size or the mandate.

$\psi > 0$ (**amplifying**): K_N grows without bound with N . There is a finite $N_c(r)$ given by (24). Any sufficiently large flat collective destabilises.

$\psi = 0$ (**marginal**): K_N is independent of N . The loop neither amplifies nor contracts, and the fixed point neither strengthens nor weakens as the collective grows. The system is poised permanently at the boundary.

$\psi < 0$ (**contracting**): $K_N \rightarrow 0$ as $N \rightarrow \infty$. The loop gain falls below $K_c(r)$ for all N , and FP2 is unconditionally stable.

The trichotomy follows from the sign of a single number, $\psi = \log_n \lambda_0$, and that number is a property of the fast microscopic law, not of the slow mandate or of the population size. Any two collectives whose fast-layer aggregation operators share a universality class sit in the same phase - a central bank and a cortical column, for instance.

C. Caveat: the fast-subsystem fork and the marginal theta mode

Caption caveat (R3, mandatory). The full closed loop (θ, m, e) carries a line of fixed points: the integrator leg $\theta_{k+1} = \theta_k + \alpha g e_k$ contributes a marginal +1 eigenvalue along θ for any $e = 0$. The two-cycle described above is a fork of the *fast* (m, e) subsystem, evaluated at the onset point $\theta = 0$ where N_c is derived. In the full loop the two-cycle is not the asymptotic attractor: the oscillation drives a secular drift of θ , which lowers the loop gain $K = \alpha g(\theta)^2$ back toward the boundary K_c and self-quenches the cycle. The long-time attractor remains the fixed-point line. The flip bifurcation - its location, amplitude scaling, and onset precision - is real and verifiable, but it belongs to the fast subsystem. Reading the fast-fork period-2 amplitude as a long-time full-loop oscillation produces a spurious finite-window bubble, not new physics.

The self-quench has a counterpart in R5: deep autonomous loops freeze rather than diverge, the slow marginal mode absorbing the excess gain.

D. Quantitative check: worked corner

In the $\mathbb{Z}_2$ majority-vote instance ($n = 2$, $\psi = 1/2$, $\beta = 0.5$, $\alpha = 0.01$, $r = 0.3$) the analytic prediction (24) gives

$$N_c = \frac{\pi(r+1)}{2\alpha\beta^2} = 816.8. \quad (25)$$

Numerical simulation of the closed (m, e) subsystem finds the onset at $N_c^{\text{num}} = 818$, a 0.15% agreement.

The square-root onset (Fig. 5) is the finite-size scaling signature of a supercritical flip, confirming the

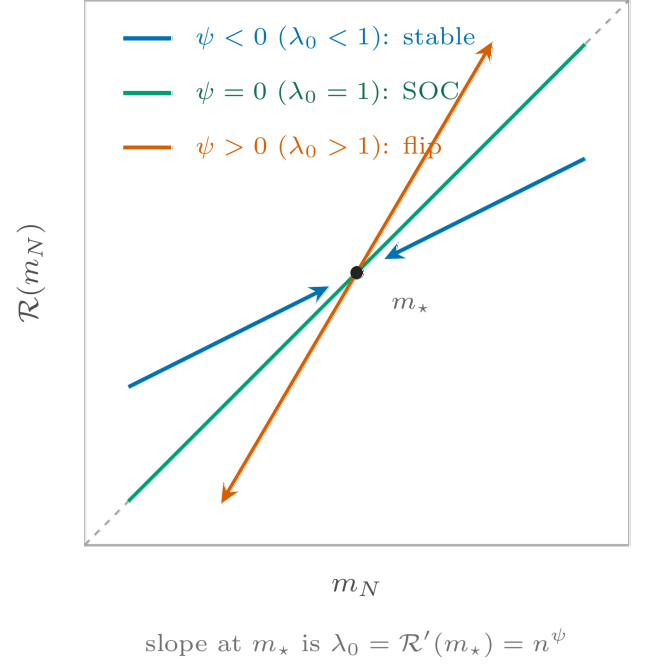


FIG. 4. (**Fig. 2b**) **Trichotomy schematic.** The three dynamical phases partitioned by the susceptibility exponent ψ . Left ($\psi < 0$): contracting - loop gain falls with N , FP2 unconditionally stable. Centre ($\psi = 0$): marginal - loop gain is N -independent, the system is poised at the phase boundary for all N . Right ($\psi > 0$): amplifying - loop gain grows with N , a finite critical size $N_c(r)$ exists above which FP2 destabilises via a flip bifurcation. The boundary $K_c(r) = r + 1$ is marked (dashed).

codimension-one character of the boundary (23). For collectives in the amplifying class, N_c is a hard wall [? ?].

VI. RESULT IV: HIERARCHICAL POSTPONEMENT AND THE PHASES OF HIERARCHY

R3 fixed the phase boundary at $K_{N_c} = K_c(r) = r + 1$. A flat loop containing $N > N_c^{(1)}$ units is unstable. The question is whether changing the architecture moves the boundary.

It does. Applying the RG decimation $\mathcal{R}$ iteratively to a hierarchy of depth L : by assumption A7, each level aggregates n units from below into one unit of identical form - the system is self-similar, so the effective gain at the top level carries the factor $\lambda_0^{-(L-1)}$ from R2:

$$K_N^{(L)} = \lambda_0^{-(L-1)} K_N^{(1)}. \quad (26)$$

The phase boundary is invariant - instability fires at the same effective gain K_c regardless of how many levels produce it. Setting $K_{N_c^{(L)}}^{(L)} = K_c$ and substituting (26)

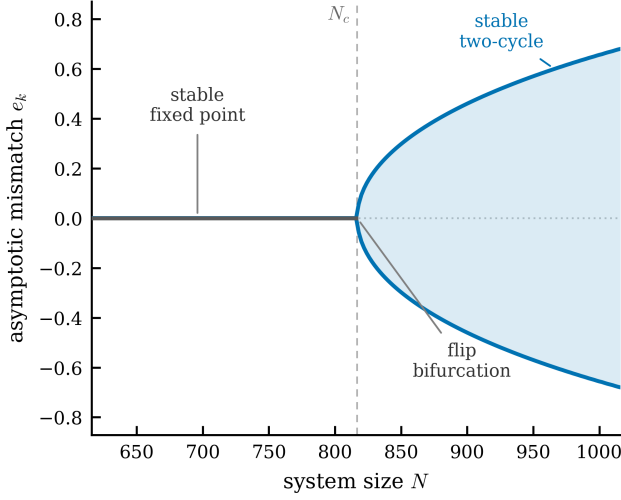


FIG. 5. (**Fig. 2c**) **Flip bifurcation data.** Amplitude of the period-2 cycle in the fast (m, e) subsystem at $\theta = 0$, swept across N (circles). Below the analytic boundary $N_c = 816.8$ (dashed vertical) the amplitude is zero to numerical precision. Above N_c the amplitude opens as $\sqrt{N - N_c}$, the universal square-root onset of a supercritical flip [? ?]. The fitted coefficient is constant just above onset, confirming the bifurcation is supercritical.

against the flat condition $K_{N_c^{(1)}}^{(1)} = K_c$ yields

$$N_c^{(L)} = N_c^{(1)} \lambda_0^{L-1}. \quad (27)$$

Each additional level multiplies the stable regime by λ_0 . Growth in population size is absorbed by growth in depth, at the fixed exchange rate λ_0 .

Inverting (27) gives the minimum depth required to hold a system of size N below the phase boundary:

$$L_{\min}(N) \sim \log_{\lambda_0} N. \quad (28)$$

Depth is renormalisation time. Promoting L levels of aggregation amounts to running the RG flow $L - 1$ steps before reading off the macroscopic gain, and the boundary recedes by one factor of λ_0 per step. Figure 6 shows the self-similar architecture: each node in the decimation tree is itself the adaptive loop of Fig. 1, and the arrow label on the right confirms that the decimation operator $\mathcal{R}$ groups n units and returns one of the same form. Figure 7 shows the consequence: $L_{\min}(N)$ against $\log N$ is a geometric staircase with equal $\log N$ risers and one $\times \lambda_0$ tread marked, and the simulated thresholds ride the predicted steps with error below 1.5%.

The scaling (28) has a second, independent derivation. Radner showed that bounded parallel information-processing capacity in an organisation enforces a minimum hierarchical depth of $L_{\min} \sim \log N$ from the economics side [? ? ?]. The present analysis reaches the same scaling by a distinct route: the requirement to hold the

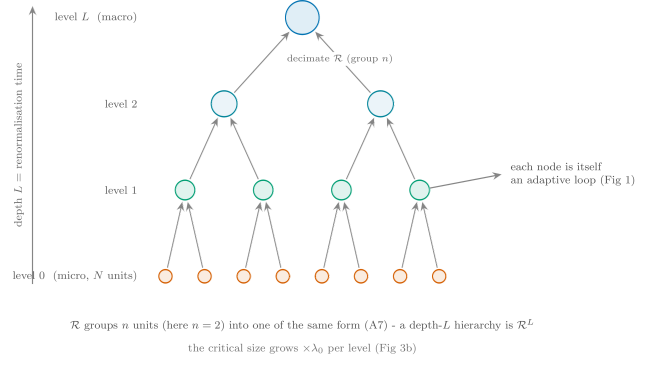


FIG. 6. **Decimation tree schematic.** A depth- L hierarchy is $\mathcal{R}^L$: each level groups n base units into one effective unit of the same form (assumption A7). The vertical axis is renormalisation time (depth L). The annotation on the right confirms each node is itself the adaptive loop of Fig. 1. The critical size grows by $\times \lambda_0$ at each level, connecting directly to Fig. 7.

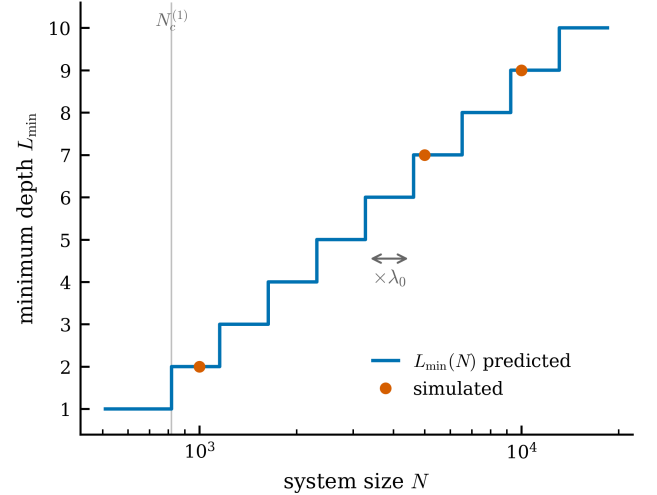


FIG. 7. **Depth staircase (R4).** Minimum stable depth $L_{\min}(N)$ plotted against $\log N$. Risers are equal in $\log N$, and the $\times \lambda_0$ span of one tread is marked. Markers are simulated thresholds from the $\mathbb{Z}_2$ majority-vote worked corner ($n = 2$, $\lambda_0 = \sqrt{2}$) - they ride the predicted treads with error below 1.5%. Read: growth demands depth, at the fixed exchange rate λ_0 .

loop gain below the flip bifurcation threshold. Two mechanisms - span-of-control limits on information processing, and susceptibility-driven feedback instability - produce the same number. The coefficient $\lambda_0 = n^\psi$ traces back to the RG eigenvalue. Ashby, Beer, and Simon identified the structural demand [? ? ?]. This derivation fixes its prefactor.

Equation (27) converts a hard phase boundary into a logarithmic depth requirement. It is also the lower wall of the viable window derived in R5: the minimum depth below which no architecture can hold a system of size N

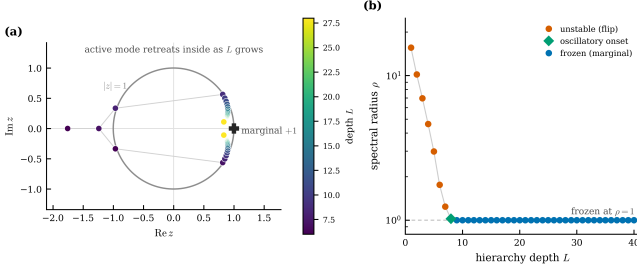


FIG. 8. **Deep loops freeze rather than destabilise.** (a) Active loop eigenvalue as a function of depth L (colour), at fixed $N > N_c^{(1)}$. Shallow loops are unstable (eigenvalue outside the unit circle). as L grows the eigenvalue retreats inside and approaches $+1$ without crossing it. (b) Spectral radius $\rho(L)$ locks onto $\rho = 1^-$ at and above $L_{\min}(N)$ and does not re-leave. There is no dynamical upper wall. The boundary that closes the viable window must therefore be energetic.

stable.

VII. RESULT V: RECONFIGURATION ENERGY AND THE VIABLE WINDOW

R4 showed that adding hierarchy levels postpones the flip boundary: $N_c^{(L)}$ grows as λ_0^{L-1} , and no finite depth makes the hierarchy itself dynamically unstable. The question is what closes the window from above.

A. The negative result: deep loops freeze

The candidate upper wall is hierarchical latency: does accumulated delay at large L return the loop to the unstable regime?

It does not. As depth grows at fixed $N > N_c^{(1)}$, the spectral radius $\rho(L)$ of the return map does not diverge. It migrates monotonically toward $\rho = 1^-$, approaching marginal stability from below (Fig. 8, panel b). At and above $L_{\min}(N)$ the active eigenvalue locks onto the unit circle and does not re-leave: the loop is frozen, not unstable (Fig. 8, panel a shows the eigenvalue track in the complex plane).

The mechanism is the same decimation that underlies R2 and R4. The map $\mathcal{R}$ attenuates the order-parameter response by $c = \lambda_0^{-1/2}$ per level (R2, A7). Each additional level reduces the effective loop gain faster than the latency penalty can raise the spectral radius. The loop holds, but it no longer moves.

B. The upper boundary is energetic

Because latency cannot produce a dynamical upper wall, the boundary that closes the viable phase must come from the cost of reconfiguration.

Model bottom-up reconfiguration as a cascade: a control input is applied at the base layer and propagates upward through L levels, each attenuating the signal by $c = \lambda_0^{-1/2}$ (A7). The state-space representation is a chain of scalar integrators with per-level gain c . The minimum energy required to move the top-layer mandate by one unit from the base, over a horizon of H update steps, is read from the reachability Gramian W_H :

$$\mathcal{E}_{\min}(L) = (e_L^\top W_H e_L)^{-1} = \lambda_0^{L-1}, \quad H \geq L. \quad (29)$$

For $H < L$ the top component of W_H vanishes: the apex is unreachable from the base in fewer than L steps, and reconfiguration cost is infinite [? ? ? ?].

No new constant enters Eq. (29). The $\lambda_0 = n^\psi$ that controls the stability gain in R4 also controls the reconfiguration cost here. Depth postpones instability at rate λ_0^{L-1} and incurs rigidity at the same rate. One eigenvalue runs in both directions simultaneously.

C. Closing the viable window

Fix a maximum available reconfiguration budget $\mathcal{E}_{\max}$ and a horizon H . The depth ceiling is

$$\Lambda = \min(H, 1 + \log_{\lambda_0} \mathcal{E}_{\max}). \quad (30)$$

The stability floor from R4 and the energy ceiling close the viable window:

$$L_{\min}(N) \leq L \leq \Lambda, \quad (31)$$

which is non-empty if and only if $N \leq N^*$, where

$$N^* = N_c^{(1)} \lambda_0^{\Lambda-1}. \quad (32)$$

N^* is the maximal governable size: the largest system that a single adaptive hierarchy of bounded depth can hold stable while remaining reconfigurable (Fig. 9). For $N > N^*$ no depth L satisfies both constraints.

D. Relation to the conservation phase

The wedge geometry has a direct counterpart in the adaptive cycle literature [? ?]. The K-phase (conservation or rigidity trap) of the Holling cycle corresponds to $L > \Lambda$: structure is sufficient to suppress instability, but reconfiguration cost exceeds the available budget. A K-phase system collapsing under external shock [? ?] is, in the present model, a forced reconfiguration that exceeds $\mathcal{E}_{\max}$: dynamically stable, but the control energy to reach the apex is exhausted.

E. Summary

R5 has two claims. First, the negative result: deep loops freeze ($\rho \rightarrow 1^-$), so the upper boundary of the viable phase is not dynamical. Second, the energetic result:

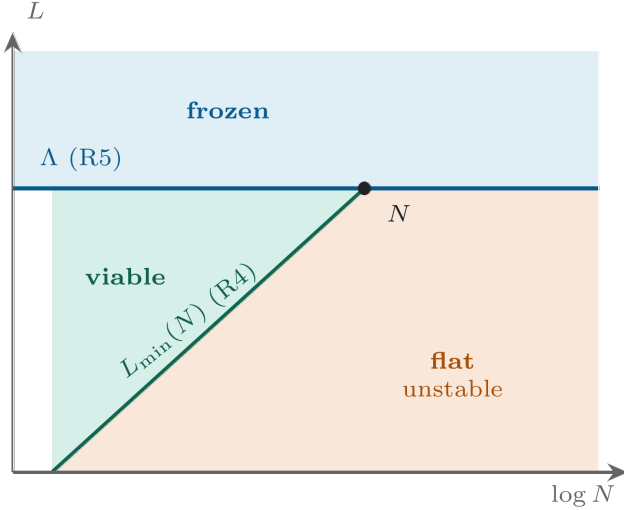


FIG. 9. **The viable window closes at N^* .** The viable phase (shaded wedge) in the $(\log N, L)$ plane, bounded below by the gain-instability floor $L_{\min}(N)$ from R4 and above by the reconfiguration ceiling Λ from R5. The two walls meet at the apex N^* . For $N > N^*$ no L satisfies both constraints simultaneously. The walls are priced by the same λ_0 .

FIG. 10. **Reconfiguration energy (data, pending).** Minimum reconfiguration energy $\mathcal{E}_{\min}(L)$ versus L , measured by two independent routes: direct inversion of the reachability Gramian (circles) and the closed-form λ_0^{L-1} (solid line). Agreement confirms that no new constant enters the upper boundary. *Note: panel pending final numerical runs.*

bottom-up reconfiguration costs $\mathcal{E}_{\min}(L) = \lambda_0^{L-1}$, fixing a depth ceiling Λ that intersects the stability floor to close the window. The same λ_0 prices both walls. Together with R1–R4, this completes the normal-form phase diagram: the adaptive loop is viable - neither unstable nor frozen - only inside the wedge $L_{\min}(N) \leq L \leq \Lambda$.

VIII. UNIVERSALITY CLASSES AND CROSS-DISCIPLINARY INSTANCES

The phase structure derived in Sections IV and ?? depends on microscopic details only through the pair (Γ, ψ) and the codimension of the instability bifurcation. This reduction is the content of a classification theorem.

Theorem 1 (Universality classification). *Let $\mathcal{S}$ be any adaptive multiscale system satisfying assumptions A1–A7. Its dynamical phase class is determined entirely by the triple (Γ, ψ, κ) , where Γ is the symmetry group of the order parameter, $\psi = \log_n \lambda_0$ is the susceptibility exponent (R2), and $\kappa \in \mathbb{Z}_{\geq 0}$ is the codimension of the bifurcation at the phase boundary. Two systems with identical (Γ, ψ, κ) share the same reduced return map up to rescaling of the*

mismatch variable, and therefore the same qualitative fate under growth in N .

Three classes follow: $\psi > 0$ (amplifying) - finite critical size N_c , flip bifurcation at the boundary. $\psi = 0$ (marginal) - self-organised criticality, gain K_N scale-invariant in N . $\psi < 0$ (contracting) - loop gain vanishes, unconditional stability at all N .

A. The worked member: $(\mathbb{Z}_2, \frac{1}{2}, \text{flip})$

The majority-vote instance has binary symmetry $\Gamma = \mathbb{Z}_2$ and group size $n = 2$, placing it in the amplifying class with $\psi = \frac{1}{2}$. The susceptibility obeys $\chi_N = AN^{1/2}$ (R2), the phase boundary is at $N_c(r) = \pi(r+1)/(2\alpha\beta^2) \approx 816.8$ in the worked corner (R3), and the transition is a codimension-one flip bifurcation of the fast (m, e) subsystem at $\theta = 0$. This member is the RG universality class of any egalitarian threshold aggregation: every system whose decimation $\mathcal{R}$ is a majority rule on $n = 2$ agents, regardless of physical substrate, has $\psi = 1/2$ and $\sqrt{N}$ growth of loop gain.

Table IV maps each domain to the loop variables (θ, h, m, T) and assigns a provisional universality class. The classification is structural: it rests on the form of the microscopic aggregation rule, not on disciplinary convention. A central bank and a recommender platform occupy the same row not because they resemble one another, but because both close a loop in which a slow mandate is updated from a delayed, finite-sample measurement of a fast population that the mandate itself drives - and both use something close to threshold aggregation. Whether the precise ψ values hold awaits measurement of the decimation eigenvalue λ_0 in each domain (R2).

B. The Condorcet reversal

The Condorcet jury theorem states that under independence, a majority vote on a binary question achieves accuracy that increases monotonically with N [? ?]. This is a local epistemic virtue. R2 states that the same statistics generate a loop gain $K_N = \alpha\chi_N g \sim \alpha AgN^\psi$. For $\psi = 1/2$ the gain grows without bound as the population grows. The accuracy and the gain are driven by the same object: the susceptibility χ_N , which is the collective sensitivity of the order parameter to the field h . Raising N improves the vote's epistemic resolution and simultaneously raises the loop gain toward the phase boundary $K_c(r) = r + 1$ (R3). There is no configuration of N that achieves Condorcet accuracy without also moving closer to the flip bifurcation.

A local epistemic virtue becomes a global dynamical liability when the same aggregation is embedded in a closed adaptive loop.

Formally: let a democratic collective grow from N to $N' > N$ while holding α, g, r , and the aggregation rule

TABLE IV. Cross-disciplinary instances of the adaptive normal form. Each domain realises the four-leg loop $\theta \rightarrow h \rightarrow \rho_h \rightarrow m \rightarrow \theta$. The Γ , ψ columns record the universality class assignment. Entries labelled (*PROVISIONAL*) require independent verification of the microscopic aggregation rule before ψ is committed. Each entry is an assignment, not an illustration: the loop structure and aggregation rule determine Γ and ψ before the domain label is read. They are not empirical measurements of ψ .

Domain	Mandate θ	Field h	Order parameter
Central bank / inflation expectations	Public inflation expectation, wage demands	Central bank rate, monetary regime	Realised inflation
Democracy ^a	Public mandate, voted preference	Enacted policy trajectory	Experienced policy
Recommender platforms ^b	User preference profile, expected engagement	Algorithm weights, content distribution	Behavioural signals
Organisations	Divisional KPIs, target expectations	Operational state, supply chain	Target-performance
Ecology ^c	Adaptive trait distribution	Carrying capacity, niche structure	Mean fitness

[†]Continuous proportional averaging places these domains in the marginal ($\psi = 0$) class. The loop gain is $O(1)$ in N and the system does not destabilise by growth alone.

^a Threshold-based mandate aggregation (majority vote) places this domain in the amplifying class. See worked member above.

^b Performative prediction [?] and model collapse [?] are the instability signatures.

^c Continuous trait adaptation lacks a binary threshold. The aggregation maps to the marginal class. Niche construction after [?].

fixed. Condorcet accuracy increases (provisionally, under the independence assumption [?]). Simultaneously, $K_{N'} > K_N$, and the system moves closer to the flip boundary. The Condorcet theorem counts only the epistemic channel. The loop carries a dynamical channel the theorem does not see. Opinion dynamics and threshold effects [?????] introduce correlations among agents that reduce the effective independence and can suppress the Condorcet accuracy, yet the loop gain is set by the *realised* susceptibility, not the i.i.d. value. Empirical correlated populations may reach the phase boundary at smaller N than the independent estimate predicts.

The thermostatic reading of democracy [??] and the adaptive-expectations framing of inflation [??] each describe a loop in which the population responds to a mismatch, not a static preference. The loop-gain analysis applies directly to both. What the normal form adds is the classification: whether a given domain sits in the amplifying, marginal, or contracting class depends on ψ , the exponent of its aggregation rule - a single number that determines how the loop gain scales with population size, and therefore which phase is reachable.

IX. DISCUSSION

R1–R5 share a single eigenvalue. The renormalisation fixed point FP3 is characterised by $\lambda_0 = \mathcal{R}'(m_*)$, the linearised slope of the decimation map at its self-similar fixed point. The susceptibility obeys $\chi_N \sim AN^\psi$ with $\psi = \log_n \lambda_0$ (R2). That power law feeds directly into the loop gain $K_N = \alpha \chi_N g$ that governs stability of the slow-fast fixed point FP2 (R1). The phase boundary $K_N = K_c(r) = r + 1$ is not a free parameter of the model - it is the dynamical shadow of the renormalisation spectrum. R5 closes the picture from above: each level of hierarchy postpones instability to $N_c^{(L)} = N_c^{(1)} \lambda_0^{L-1}$ (R4), but the

same λ_0 prices the energy required to reconfigure that hierarchy as $\mathcal{E}_{\min}(L) = \lambda_0^{L-1}$. One number is the exchange rate between stability gain and reconfiguration rigidity. The phase diagram is a closed wedge in (N, L) space, non-empty up to $N^* = N_c^{(1)} \lambda_0^{\Lambda-1}$, with both boundaries receding at the same rate.

Whether the order parameter is realised inflation tracked against a household-expectation mandate, an aggregated vote share tracked against a coalition position, or a population-level observable tracked against a policy rate, the phase boundary is the same formal object: $K_N = r + 1$. The specific system enters only through α , g , and A . The exponent ψ and the bifurcation type (flip, for $\mathbb{Z}_2$ -symmetric loops) are properties of the renormalisation fixed point, not of the domain.

Several extensions lie outside the settled results. Assumption A4 restricts the slow law to gradient descent on the variational principle $F[\theta]$. Lifting this to the general non-gradient (solenoidal) case opens an A4 extension in which θ can orbit a limit cycle rather than settle at a fixed point, admitting limit-cycle attractors in parameter regimes where R3 predicts only fixed-point convergence. Assumption A3 invokes the central limit theorem to establish the $N^{-1/2}$ noise floor. The heavy-tailed (generalised central limit theorem, α -stable) extension could alter the susceptibility exponent ψ and shift the trichotomy boundaries in ways the present derivation does not reach. The multidimensional case - θ a vector, F a landscape with multiple basins - is the natural next step once the scalar normal form is published. The single-component analysis here is the necessary base case. Two instance mappings - transformer attention as a depth- L renormalisation loop, and reservoir computers as the $\psi \rightarrow 0$ marginal case - are deferred to preserve scope.

Empirical access runs through the stability ratio $\Xi_t = K_{\text{eff}}(t)/K_c(T, \tau_t)$, which converts the theoretical phase boundary into a time-series observable computable from

structural data alone. Two domains offer the cleanest measurement sites because their aggregation mechanism most closely realises the Γ -equivariant loop of the normal form. In inflation-expectations dynamics, θ_t is the surveyed household expectation, the order parameter m_t is realised inflation, and the mismatch e_t is the forecast error. Shifts in media salience or co-ordination events move K_{eff} detectably against the stable background [? ?]. In electoral polling, the mandate θ is the coalition position revised between cycles and the order parameter is the aggregated vote share. The flip bifurcation of R3 predicts alternating two-cycles in the mismatch series as $\Xi_t \rightarrow 1$, a signature distinct from the variance divergence expected at a continuous transition [? ? ? ?]. Systematic validation against macroeconomic survey data and electoral time series is deferred to a companion empirical study.

X. CONCLUSION

The adaptive phase transition is the shadow of the renormalisation fixed point FP3. The susceptibility $\chi_N \sim AN^\psi$ grows with system size at a rate set by the linearised RG eigenvalue λ_0 , and this growth drives the loop gain $K_N = \alpha\chi_N g$ through the flip boundary $K_c = r + 1$ at a finite critical size N_c (R1). The exponent $\psi = \log_n \lambda_0$ is the only quantity that classifies which dynamical phase a system occupies. $\psi > 0$: instability is inevitable at large enough N . $\psi < 0$: the loop is unconditionally stable.

$\psi = 0$: self-organised criticality (R2, R3).

A system in the amplifying class ($\psi > 0$) can postpone the flip bifurcation by adding levels of hierarchy. Each level scales χ_N down by λ_0 , moving the boundary to $N_c^{(L)} = N_c^{(1)} \lambda_0^{L-1}$. To remain stable at size N , a system needs at least $L_{\min} \sim \log_{\lambda_0} N$ levels - depth is the only lever (R4).

The viable window $L_{\min}(N) \leq L \leq \Lambda$ has two walls. Deep autonomous loops do not lose stability - they freeze. The spectral radius $\rho(L)$ locks onto 1 once $L \geq L_{\min}$ and does not leave, so there is no dynamical instability from above. The ceiling is energetic: the minimum energy required to reconfigure the top of an L -level hierarchy from its base is $\mathcal{E}_{\min}(L) = \lambda_0^{L-1}$, set by the same eigenvalue that determines the stability floor. A finite energy budget $\mathcal{E}_{\max}$ and a finite planning horizon H together fix a depth ceiling $\Lambda = \min(H, 1 + \log_{\lambda_0} \mathcal{E}_{\max})$. The window closes at the maximal governable size $N^* = N_c^{(1)} \lambda_0^{\Lambda-1}$. Both walls are priced by λ_0 simultaneously: the window is a consequence of the renormalisation spectrum, not a coincidence of the parameters (R4, R5).

A central bank steering inflation expectations and a cortical column integrating sensory streams across laminar depth are the same loop: ψ classifies each, and Λ is the ceiling past which no amount of additional structure restores reconfigurability. The normal form is not an analogy across these domains - it is the object they are all instances of. What remains open is the measurement: the decimation eigenvalue λ_0 has not been extracted from any of the candidate empirical systems. That measurement is the next step.